BIS629. Advanced methods for implementation and prevention science HW#4, Fall 2026

Due: Tuesday, November 17 at 3PM

When using AI to solve these problems, please be sure to include an appendix with your prompts, output and your method(s) of verification. The method of verification should be shown in sufficient detail so that we can evaluate its validity.

This problem is based, in part, on "MaxART_HW4_2026.RData" which was uploaded to Homework 4 folder in Canvas. The MaxART_HW4_2026.RData" has a total of 6 variables, including

- "id": is the ID for each participant
- "Y": a binary outcome (1 if CD4 count >= 350 cells/mm^3 at 180 days post ART initiation, otherwise 0)
- "treatment": 0, SoC; 1, EAAA
- "clinc": name for each cluster (001--014), a total of 14 clusters
- "stop": stop taking medication. This variable is not relevant to this homework, please ignore it.
- "step": at which step this clinic transits to EAAA.

NOTICE: please note that "step" indicates the step when the clinic transits to EAAA. Therefore, in each clinic, some participants received treatment 0 at enrollment (before their clinic transitioned to EAAA) and the others received treatment 1 at enrollment (after their clinic transitioned to EAAA). To avoid survivor bias, we maintain the initial treatment assignment for each participant. Otherwise, after a clinic transitions, only survivors of standard of care will continue on to the intervention, disturbing the initial randomization and possibly making the participant groups in standard of care and intervention not comparable. *For all power calculations, for comparability across software options, use the identity link function. Only swdpwr supports link functions other than the identity. Although PASS allows effect sizes to be input and output as odds ratios and risk ratios, it converts them to proportions and bases calculations on the identity link as in Hussey and Hughes.*

1. How many steps does MaxART have? Show the schematic diagram that describes the design of MaxART.

2. Calculate the power of MaxART to detect an intervention effect corresponding to an RR of 0.80 with all other design features the same as those in the data. Use the proportion in the control group as observed in the MaxART data. Use the definition of the intra-class correlation coefficient given in this course.

a) Using swdpwr, first assume the marginal and conditional models without time effects. Please specify all input parameters needed.

b) Next, use PASS, the Hussey and Hughes spreadsheet, and swdpwr, assuming the marginal model allowing for a time effect. Assume small, medium and large time effects (e.g. 1%, 25%, 100% of the baseline outcome rate from start to end of follow-up). For example, if the baseline outcome rate is 80% and the time effect is assumed to be a 1% increase in the outcome rate from baseline to end of follow-up under control conditions, the expected outcome rate in the control group at the end of the study period will be 0.80 + 0.80*0.01. Please specify all input parameters needed, if different from part a) above.

c) Please create a well-labeled table summarizing your results. Please discuss how the results differ using three software options, with and without considering small, moderate and strong time effects?

3. Now calculate the power of MaxART, assuming the same total sample size, but with 12 clusters, and with 3 steps, with all parameters other than the step size and average number of participants the same as in question 2.

a) Show the schematic diagram that describes this design

Calculate the power of this design assuming

b) the marginal model using PASS, Hussey and Hughes spreadsheet, and the swdpwr package with time effect equal to 1% baseline outcome rate from the start to the end of follow up, and with no time effect

c) the conditional model using swdpwer package with the time effect equals to 1% baseline outcome rate from the start to the end of follow up, and with no time effect.

Create a table with your results and briefly summarize your findings in words.

4. Now repeat question 3 but calculate the power of MaxART to detect an OR corresponding to a RR of 0.80.

a) What is this OR?

b) Add these results to the table in question 3, and briefly summarize your further findings in words. How do the results change when we define the target parameter of interest as the OR instead of the RR?